

Environmental Thermal Chamber Validation Test Report

Climatic, Thermal Shock, and Temperature Sensor Qualification per ISO 16750-4

1. Cell Temperature Sensor Chamber Measurements

- Test Case TC-ENV-003: Cell temperature measurement verified across cold and warm chambers from -20.0°C to +70.0°C with accuracy ±0.5°C.
- Sensor calibration profile: NTC thermistor polynomial coefficient verification.

2. Thermal Shock Fast Transition Endurance Test

- Test Case TC-ENV-094: Thermal shock chamber test completed 250 cycles of 500 total cycles with zero solder joint cracking. Remaining units passed functional testing.
- Chamber transition time: 22 seconds between -40°C basket and +85°C basket.